D. Blackslex and Penguin Civilization

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1300 | Tags: bitmasks, constructive algorithms, greedy, math | https://codeforces.com/problemset/problem/2179/D

Penguins are civilized creatures that communicate using permutations. Blackslex, as a penguin researcher, must study their means of communication.

For a given integer n, consider permutations[□] p of the array $[0, 1, \ldots, 2^n - 1]$. Define $S(p) = \sum_{i=0}^{2^n-1} \text{popcount}(p_0 \mathbin{\&} p_1 \mathbin{\&} \ldots \mathbin{\&} p_i)$, where popcount($z$) is the number of 1-bits in the binary representation of z (for instance, popcount(5) = 2 because $5 = 101_2$ has two 1-bits in the binary representation), and & denotes the bitwise AND operation. .

A permutation is considered **sacred** if it maximizes $S(p)$. Find the **lexicographically minimal**[†] sacred permutation.

[□]A permutation of length n is an array consisting of n distinct integers from 1 to n in arbitrary order. For example, [2,3,1,5,4] is a permutation, but [1,2,2] is not a permutation (2 appears twice in the array), and [1,3,4] is also not a permutation (n=3 but there is 4 in the array).

[†]An array a is lexicographically smaller than an array b of the same size if and only if the following holds:

- in the first position where a and b differ, the array a has a smaller element than the corresponding element in b.

Input

The first line contains a single integer t ($1 \le t \le 16$) — the number of test cases.

Each test case contains a single integer n ($1 \le n \le 16$).

It is guaranteed that the sum of 2^n over all test cases does not exceed 2^{16}.

Output

For each test case, output 2^n integers $p_0, p_1, \ldots, p_{2^n-1}$ — the required permutation.

Example

Input
2 1 2

Output
1 0 3 1 0 2

Note

For the first test case, there are two possible permutations.

- $p = [0, 1]$, $S(p) = 0$
- $p = [1, 0]$, $S(p) = 1$

For the second test case, $S([3, 1, 0, 2]) = 3$ is sacred. There are other permutations p that are sacred, such as $p = [3, 2, 0, 1]$, but those are not lexicographically minimal.